Python Tkinter Tutorial

Student Calculator aur Greeting App banana seekhein

Shuruwaat karne walon ke liye — Qadam ba Qadam Guide

Aap kya banayenge:

Is tutorial mein aap ek Tkinter GUI application banayenge jo user ka naam lekar usse greet karegi, aur saath hi addition aur subtraction bhi karegi. Is tutorial ke baad aap samjhenge ke libraries kaise import hoti hain, functions kaise banate hain, window kaise create hoti hai, aur interactive widgets kaise add kiye jaate hain.

Section 1 — Zaroori Libraries Import Karna

Har Python program imports se shuru hota hai. Yahan hum Python ki standard library se apne zaroori tools load karte hain:

```
import tkinter as tk
from tkinter import messagebox
```

Har import ki wazahat:

- **`tkinter`**: Yeh Python ke saath built-in GUI framework hai. Isse hum Window, Button, Label, Entry, aur doosre UI elements banate hain.
- **`messagebox`**: Yeh Tkinter ka sub-module hai jo pop-up dialogs dikhata hai — jaise warnings, errors, aur informational alerts.

Section 2 — Functions Define Karna

Python mein functions reusable logic ko group karte hain. Hamare application ko teen functions chahiye: ek user ko greet karne ke liye, aur do calculations ke liye.

2.1 greet() — Greeting Function

Jab user 'Greet Me' button click kare, yeh function naam padh kar ek personalized message dikhata hai:

```
def greet():
```

```
    name = entry_name.get()          # Name field se text padho
    if name == "":                   # Agar input khali ho
        messagebox.showwarning(
            "Warning",
            "Please enter your name!"
        )
    else:
        label_greet.config(           # Greeting label update karo
            text="Hello, " + name + "!"
        )
```

Istemal kiye gaye key concepts:

- **`entry_name.get()`**: Tkinter Entry widget se current string padhta hai.
- **`messagebox.showwarning()`**: Title aur message ke saath warning pop-up dikhata hai.
- **`label.config(text=...)`**: Label widget ka displayed text dynamically update karta hai.

2.2 add_numbers() — Addition Function

Yeh function user ke daale hue do numbers leke unhe add karta hai aur result dikhata hai. try/except block galat (non-numeric) input ko handle karta hai:

```
def add_numbers():
    try:
        num1 = float(entry_num1.get())    # Pehla input decimal mein convert karo
        num2 = float(entry_num2.get())    # Doosra input decimal mein convert karo
        result = num1 + num2
        label_result.config(text="Result: " + str(result))
    except ValueError:
        messagebox.showerror(
            "Error",
            "Please enter valid numbers!"
        )
```

Key concepts:

- **`float()`**: String ko floating-point (decimal) number mein convert karta hai.
- **`try / except ValueError`**: Woh error pakadta hai jo float() ko non-numeric text milne par aati hai — program crash nahi hota.
- **`str()`**: Numeric result ko display ke liye wapas string mein convert karta hai.

2.3 subtract_numbers() — Subtraction Function

Yeh function bilkul add_numbers() ki tarah kaam karta hai, bas addition ki jagah subtraction karta hai:

```
def subtract_numbers():
    try:
        num1 = float(entry_num1.get())
        num2 = float(entry_num2.get())
        result = num1 - num2
        label_result.config(text="Result: " + str(result))
    except ValueError:
        messagebox.showerror(
            "Error",
            "Please enter valid numbers!"
        )
```

> **Mashwarah:**
> Subtraction exactly wahi error-handling pattern follow karta hai jo addition mein hai. Ek advance exercise ke taur par, aap dono functions ko ek mein merge kar sakte hain jo operator as argument leta ho — jab yeh tutorial mukammal karen tab zaroor try karein!

Section 3 — Main Window Banana

Koi bhi widget add karne se pehle hum application window create karte hain aur uski properties set karte hain:

```
root = tk.Tk()                          # Main application window create karo
root.title("Student Calculator App")    # Title bar ka text set karo
root.geometry("400x450")                # Window size: width x height (pixels)
root.config(bg="#EBF5FB")               # Saaf light-blue background colour
root.resizable(False, False)            # User ko window resize karne se rokna
```

Properties ki wazahat:

- **`tk.Tk()`**: Tkinter library initialize karta hai aur root (main) window object banata hai.
- **`root.title()`**: OS title bar mein dikhne wala text set karta hai.
- **`root.geometry()`**: "WIDTHxHEIGHT" format mein string leta hai aur window ko un pixel dimensions par fix karta hai.
- **`root.config(bg=...)`**: Window canvas ka background colour apply karta hai.
- **`root.resizable(False, False)`**: Window size lock kar deta hai taake layout kabhi break na ho.

Section 4 — Naam aur Greeting Section Banana

Ab hum window mein widgets add karte hain. Widgets ko pack() geometry manager se vertically stack kiya jata hai:

```
# --- Greeting Section ---
tk.Label(root, text="Enter Your Name:",
         bg="#EBF5FB", font=("Arial", 11)).pack(pady=6)

entry_name = tk.Entry(root, width=30, font=("Arial", 11))
entry_name.pack(pady=6)

tk.Button(root, text="Greet Me",
          command=greet,
          bg="#2E75B6", fg="white",
          font=("Arial", 11, "bold"),
          relief="flat", padx=10).pack(pady=6)

label_greet = tk.Label(root, text="", bg="#EBF5FB",
                       font=("Arial", 12, "italic"), fg="#1F4E79")
label_greet.pack(pady=4)
```

Widgets ka jaiza:

- **`tk.Label()`**: Window ke andar read-only text ya images dikhata hai.
- **`tk.Entry()`**: Single-line text input field jahan user type karta hai.
- **`tk.Button()`**: Clickable button; command parameter ise function se link karta hai.
- **`.pack(pady=N)`**: Automatic stacking se widget position karta hai aur N pixels vertical padding add karta hai.

Section 5 — Calculator Section Banana

Hum do number inputs aur do operation buttons add karte hain. Wahi entry widgets dono add aur subtract functions share karte hain:

```
# --- Calculator Section ---
tk.Label(root, text="Enter First Number:",
         bg="#EBF5FB", font=("Arial", 11)).pack(pady=6)
entry_num1 = tk.Entry(root, width=30, font=("Arial", 11))
entry_num1.pack(pady=6)

tk.Label(root, text="Enter Second Number:",
         bg="#EBF5FB", font=("Arial", 11)).pack(pady=6)
entry_num2 = tk.Entry(root, width=30, font=("Arial", 11))
entry_num2.pack(pady=6)
```

```
# Operation Buttons (side by side)
btn_frame = tk.Frame(root, bg="#EBF5FB")
btn_frame.pack(pady=8)

tk.Button(btn_frame, text="  Add  ", command=add_numbers,
          bg="#27AE60", fg="white",
          font=("Arial", 11, "bold"), relief="flat").pack(side="left", padx=8)

tk.Button(btn_frame, text="Subtract", command=subtract_numbers,
          bg="#E74C3C", fg="white",
          font=("Arial", 11, "bold"), relief="flat").pack(side="left", padx=8)

label_result = tk.Label(root, text="Result: -",
                        bg="#EBF5FB", font=("Arial", 13, "bold"), fg="#1F4E79")
label_result.pack(pady=10)
```

> **Layout Mashwarah:**
> tk.Frame ko container ke taur par use karne se hum pack(side='left') ke zariye multiple buttons side by side rakh sakte hain — jo single column layout mein possible nahi hota.

Section 6 — Main Event Loop Shuru Karna

Har Tkinter program ki aakhri line control Tkinter ke event loop ko de deti hai:

```
root.mainloop()    # Window khuli rakhta hai aur user interactions handle karta
hai
```

mainloop() kya karta hai:

- Yeh lagatar events sunta rehta hai: button clicks, key presses, window close signals, wagera.
- Jab user koi event trigger kare to suitable callback function call karta hai (jaise greet, add_numbers).
- Jab user window band kare to program cleanly exit ho jata hai.

Mukammal Program — Pura Source Code

Yeh Student Calculator App ka mukammal, production-ready version hai:

```
import tkinter as tk
from tkinter import messagebox
```

```
# -- Functions ---------------------------------------------------

def greet():
    name = entry_name.get()
    if name == "":
        messagebox.showwarning("Warning", "Please enter your name!")
    else:
        label_greet.config(text=f"Hello, {name}! App mein khush aamdeed.")


def add_numbers():
    try:
        result = float(entry_num1.get()) + float(entry_num2.get())
        label_result.config(text=f"Result: {result}")
    except ValueError:
        messagebox.showerror("Error", "Please enter valid numbers!")



def subtract_numbers():
    try:
        result = float(entry_num1.get()) - float(entry_num2.get())
        label_result.config(text=f"Result: {result}")
    except ValueError:
        messagebox.showerror("Error", "Please enter valid numbers!")



# -- Main Window -------------------------------------------------

root = tk.Tk()
root.title("Student Calculator App")
root.geometry("400x450")
root.config(bg="#EBF5FB")
root.resizable(False, False)

# -- Greeting Section --------------------------------------------

tk.Label(root, text="Enter Your Name:", bg="#EBF5FB",
         font=("Arial", 11)).pack(pady=6)
entry_name = tk.Entry(root, width=30, font=("Arial", 11))
entry_name.pack(pady=6)
tk.Button(root, text="Greet Me", command=greet,
```

```
           bg="#2E75B6", fg="white",
           font=("Arial", 11, "bold"), relief="flat", padx=10).pack(pady=6)
label_greet = tk.Label(root, text="", bg="#EBF5FB",
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label_greet.pack(pady=4)

# -- Calculator Section ----------------------------------------

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label_result.pack(pady=10)

# -- Application Shuru Karo -------------------------------------

root.mainloop()
```

Khulasa — Ahem Concepts Ek Nazar Mein

Concept	Python Element	Kaam (Maqsad)
GUI Framework	`import tkinter as tk`	Windows aur widgets banana
Pop-up Dialogs	`from tkinter import messagebox`	Warning / error alerts dikhana
Input Widget	`tk.Entry()`	User ka type kiya text pakadna

Display Widget	`tk.Label()`	Static ya dynamic text dikhana
Click Action	`tk.Button(command=fn)`	Click par function trigger karna
Error Handling	`try / except ValueError`	Galat input par crash rokna
Type Conversion	`float() / str()`	String aur number ke darmiyan convert karna
Event Loop	`root.mainloop()`	App chalti aur responsive rakhna